Project Title: STEM Course Access & Enrollment Dashboard

Project Overview

This Power BI dashboard analyzes public school enrollment in mathematics and science courses across U.S. states using 2013–14 Civil Rights Data Collection data published through Data.gov. The goal of the project was to evaluate course enrollment patterns, compare participation across STEM subjects, and create an executive-style reporting dashboard that highlights trends by state, course category, and student group.

The dashboard was designed as a portfolio project to demonstrate end-to-end data analytics skills, including data cleaning, Power Query transformation, data modeling, DAX measure development, dashboard design, KPI reporting, and insight communication.

Data Source

The dataset used for this project is the 2013–14 Mathematics and Science Civil Rights Data Collection, published by the U.S. Department of Education and available through Data.gov. According to the dataset description, the 2013–14 tables are based on data collected from approximately 16,500 school districts and 96,500 schools across the United States. The files include data for schools offering mathematics and science courses and student enrollment by state. For each mathematics and science course, the source files include separate spreadsheets for total students, male students, and female students.

The broader Civil Rights Data Collection 2013–14 is a cross-sectional survey that collects data on key education and civil rights issues in public schools, including student enrollment and educational programs and services. The data is disaggregated by race/ethnicity, sex, limited English proficiency, and disability status.

Data Files Used

The dashboard combines multiple Excel workbooks from the Mathematics and Science Civil Rights Data Collection, including:

- Enrollment in Advanced Mathematics
- Enrollment in Algebra I Grades 7–8

- Enrollment in Algebra I Grades 9–10
- Enrollment in Algebra I Grades 11–12
- Enrollment in Algebra II
- Enrollment in Biology
- Enrollment in Calculus
- Enrollment in Chemistry
- Enrollment in Geometry
- Enrollment in Physics

A separate mathematics and science classes file was reviewed as a related source, but the primary dashboard pages focus on course enrollment files.

Data Preparation & Transformation

The original data was provided as multiple Excel files with similar structures but required several preparation steps before it could be used in Power BI visuals. I used Power Query to clean and consolidate the files into reporting-ready tables.

Key transformation steps included:

- Imported multiple Excel workbooks from a folder source.
- Filtered out non-enrollment files and temporary Excel files.
- Extracted course names from file names.
- Expanded workbook sheets and retained the sheet name as the Gender Group field.
- Standardized course names, including Algebra I grade-band courses.
- Removed title/header rows and renamed columns by position to handle inconsistent source headers.
- Appended all enrollment files into one combined table.
- Unpivoted demographic enrollment columns into a long-format structure.
- Created fields for Student Group, Metric Type, and Enrollment Value.
- Converted numeric fields while preserving original values for auditability.
- Handled suppressed values labeled “1 to 3” by assigning the midpoint value of 2, consistent with the source note.
- Created a separate school reporting table for number of schools and reporting-rate metrics.
- Added course category logic to classify courses as Mathematics or Science.

Data Model

The model was designed around two primary reporting tables:

Enrollment Long

Used for enrollment analysis by state, course, course category, gender group, student group, metric type, and enrollment value.

School Reporting

Used for school reporting counts and reporting-rate analysis by state, course, course category, and gender group.

This structure supports flexible filtering across course, state, gender group, course category, and data status while reducing double-counting through DAX measures.

Key DAX Measures

Several DAX measures were created to support KPI cards, rankings, and comparison visuals, including:

- Total Enrollment
- Total Schools Reporting
- Average Reporting Rate
- Courses Analyzed
- Course Enrollment Share
- Highest Enrollment Course
- Lowest Enrollment Course
- Suppressed Value Count

The main enrollment measure filters to total students, number-based values, and total gender group to prevent double-counting across demographic and gender rows.

Dashboard Pages

Page 1: Executive Overview

The first page provides a high-level summary of STEM course enrollment across states and courses. It includes KPI cards for total enrollment, school reporting, average reporting rate, and number of courses analyzed. It also includes course enrollment rankings, a state-level enrollment visual, and a state summary table.

This page was designed to simulate an executive dashboard that quickly answers:

- How many students are represented in the course enrollment data?
- Which courses have the highest reported enrollment?
- Which states have the highest enrollment totals?
- How complete is the school reporting coverage?

Page 2: Course Enrollment Comparison

The second page focuses on comparing enrollment across mathematics and science courses. It includes course ranking visuals, top and lowest enrollment course KPIs, a course category comparison, enrollment share analysis, and an insight callout.

This page was designed to answer:

- Which STEM courses have the highest and lowest reported enrollment?
- How does enrollment compare between mathematics and science courses?
- Which courses make up the largest share of reported enrollment?
- How should enrollment patterns be interpreted alongside reporting coverage?

Key Insights

The dashboard shows that enrollment is highest in foundational STEM courses such as Biology, Geometry, Algebra I, and Chemistry, while advanced courses such as Calculus and Physics show lower reported participation. This pattern may suggest that advanced STEM course access or participation is more limited than foundational course enrollment.

The analysis also highlights the importance of interpreting enrollment totals alongside reporting coverage. Since school reporting counts are aggregated across courses and states, they should be used as a coverage indicator rather than interpreted as a unique school count.

Data Limitations

This project includes several important limitations:

- The data reflects the 2013–14 school year, so it should be treated as historical rather than current.
- School reporting counts are aggregated by course and state and should not be interpreted as unique schools across the full dataset.
- Suppressed values labeled “1 to 3” were converted to a midpoint value of 2 for numeric analysis.
- The dashboard focuses on state-level course enrollment and does not include school-level or district-level drill-downs.
- Enrollment totals may vary by reporting coverage and should be interpreted alongside reporting-rate metrics.

Skills Demonstrated

This project demonstrates the following Power BI and analytics skills:

- Power BI dashboard development
- Power Query data cleaning and transformation
- Folder-based file ingestion
- Excel workbook consolidation
- Data modeling and long-format table design
- DAX measure creation
- KPI design and executive reporting
- Course/category classification
- Data quality handling and suppressed value treatment
- Interactive slicer and filter design
- Data storytelling and insight communication
- Portfolio Positioning Statement

This dashboard was created as a portfolio project using public education data to demonstrate my ability to transform multiple raw Excel workbooks into a clean Power BI reporting model, develop DAX measures, design executive-style dashboards, and communicate actionable insights from complex operational-style data.